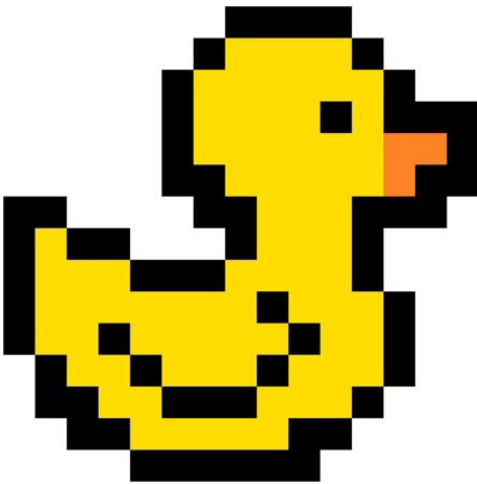


SECURITY INCIDENT REPORT

BlackCat / ALPHV Ransomware Attack

PawPressMe

Field	Value
Incident ID	INC-2026-0922-001
Date / Time	22 September 2026 07:16 – 07:32 UTC+7
Severity	● CRITICAL
Verdict	TRUE POSITIVE (Ransomware / Double Extortion)
Threat Actor	BlackCat / ALPHV (suspected)
Attack Vector	Credential Leak → SSH Brute Force → Ransomware
Affected Endpoints	paw-tom, paw-leo, paw-marie, paw-tabitha (×4 encrypted)
Cloud Impact	10.10.50.10 — IMDS token theft suspected
Data Exfiltration	Yes (3 SFTP sessions from paw-backup)
Prepared By	Jovita Kusuma
Status	Active Investigation



1. Executive Summary

On 22 September 2026, PawPressMe was struck by a full-chain BlackCat/ALPHV ransomware attack. The threat actor obtained plaintext credentials for four employees from the dark-web credential marketplace WatchingCatsLounge.thm and used them to brute-force SSH access to the Linux backup server paw-backup.

Once inside, the attacker performed local account enumeration, pivoted to the pawbackup service account, and exfiltrated backup data across three covert SFTP sessions. In parallel, an unknown cloud asset (10.10.50.10) made 22 requests to the Azure Instance Metadata Service (IMDS), consistent with OAuth2 token theft for cloud lateral movement.

The attacker then used a shared local administrator account (itadmin-tom) to create a Windows Scheduled Task named "RunRansom" across all four user endpoints and executed ransom.exe, which spawned 30+ encryption threads per host. All four endpoints are presumed encrypted.

2. Attack Timeline

Time (UTC+7)	Host / Source	Event
Pre-incident	Dark web	Employee credentials (leo, marie, tabitha, tom) leaked on WatchingCatsLounge.thm
07:16:35	paw-backup	First SSH events captured by Sysmon — attack begins
07:17:56	paw-backup	SSH brute-force from 194.50.16.198: Invalid users leo, marie, tabitha; tom is a valid account
07:18:00	paw-backup	SSH brute-force continues — Failed password events for all 4 accounts
07:18:24	paw-backup	Successful SSH login as tom (last failed attempt then session established)
07:19:30	10.10.50.10	Cloud asset initiates outbound connections to 4.208.26.197:443 and 185.199.108.133:443
07:19:31	10.10.50.10	First IMDS request: curl to 169.254.169.254 (user: tom)
07:20:45–46	10.10.50.10	IMDS burst: 22 requests total to 169.254.169.254:80 (Azure token enumeration)
07:29:43	paw-backup	Covert SFTP exfiltration — Session 1: bash -c /usr/lib/openssh/sftp-server (tom, TerminalSession 6)
07:30:51	paw-backup	Covert SFTP exfiltration — Session 2: bash -c /usr/lib/openssh/sftp-server (pawbackup)
~07:32+	paw-backup	Log suppression begins — SSH and Sysmon events cease on paw-backup
~07:52	paw-tom	schtasks RunRansom created; ransom.exe executed by itadmin-tom — 30+ encryption threads
~07:56	paw-leo	schtasks RunRansom created; ransom.exe executed — 30+ encryption threads
~08:00	paw-marie	schtasks RunRansom created; ransom.exe executed — 30+ encryption threads
~08:05	paw-tabitha	schtasks RunRansom created; ransom.exe executed — 30+ encryption threads

3. Attack Stages — Detailed Analysis

Stage 1 · Initial Access — Credential Theft & SSH Brute Force

MITRE Tactic	Technique ID	Technique Name
Initial Access	T1078 T1110.001	Valid Accounts Brute Force: Password Guessing

What happened

The attacker obtained plaintext credentials for four PawPressMe employees from the dark-web forum WatchingCatsLounge.thm before the attack. Using these, the actor launched an SSH brute-force from IP 194.50.16.198 against paw-backup. Three accounts (leo, marie, tabitha) were rejected as "Invalid user" — they do not exist on the system. The fourth account, tom, was confirmed as valid and authenticated successfully at approximately 07:18:24.

Key Evidence

- Attacker IP: 194.50.16.198 (external, not in PawPressMe IP space)
- paw-backup SSH log: "Invalid user leo/marie/tabitha from 194.50.16.198"
- "Failed password for tom from 194.50.16.198" — multiple ports (60422, 34792, 55254)
- Successful login inferred: tom activity begins at 07:18:24 with no prior valid session
- Credential source confirmed: WatchingCatsLounge.thm DarkSearch alert for pawpress.me

Stage 2 · Cloud Credential Theft — IMDS Enumeration

MITRE Tactic	Technique ID	Technique Name
Credential Access	T1552.005	Cloud Instance Metadata API

What happened

After establishing access, user "tom" (from cloud asset 10.10.50.10) used curl to query the Azure Instance Metadata Service (IMDS) endpoint at 169.254.169.254 a total of 22 times. IMDS provides OAuth2 tokens for the VM's managed identity — querying it without proper network controls allows an attacker to harvest Azure API tokens for lateral movement into cloud resources. The asset also connected to two external IPs: 4.208.26.197 and 185.199.108.133 (GitHub CDN range), suggesting tool download or data staging.

Key Evidence

- Source: 10.10.50.10 — undocumented cloud asset, not in known PawPressMe inventory
- Tool used: /usr/bin/curl (observed in Sysmon network connection events, user: tom)
- Destination: 169.254.169.254:80 — Azure IMDS endpoint hit 22 times
- External connections: 4.208.26.197:443 and 185.199.108.133:443 (GitHub CDN)
- Technique: IMDS token retrieval bypasses standard Azure IAM logging

Stage 3 · Post-Exploitation Recon — Local Account Discovery

MITRE Tactic	Technique ID	Technique Name
Discovery	T1087.001	Account Discovery: Local Account

What happened

Once logged in as tom, the attacker ran "cat /etc/passwd" from the /home/tom directory. This dumps all local accounts on the Linux system — including usernames, UIDs, home directories, and default shells — allowing the attacker to identify the pawbackup service account as a target for lateral movement. Sysmon EventID 1 captured the process creation.

Key Evidence

- Command: cat /etc/passwd
- User: tom | Working directory: /home/tom
- Process: /usr/bin/cat | PID: 9505
- Sysmon RuleName: TechniqueID=T1087.001,TechniqueName=Account Discovery: Local Account
- Process GUID: {ec24ddd5-3c16-6827-b134-0da5ee580000}

Stage 4 · Lateral Movement — Switch to pawbackup Service Account

MITRE Tactic	Technique ID	Technique Name
Lateral Movement	T1078.003	Valid Accounts: Local Accounts

What happened

After enumerating local accounts, the attacker switched from the tom user to the pawbackup service account. This account owns the backup data directory (/home/pawbackup) and has the permissions needed to read and transfer backup files. The exact su/sudo command was masked by log suppression that began shortly after, but the switch is confirmed by the SFTP exfiltration events that follow under the pawbackup identity.

Key Evidence

- tom session last confirmed: 07:29:43 (SFTP Session 1 — TerminalSession 6, LogonId 1002)
- pawbackup session confirmed: 07:30:51 (SFTP Session 2 — LogonId 1003)
- Log suppression begins ~07:32 — su event not captured
- pawbackup working directory: /home/pawbackup (confirmed in Sysmon CurrentDirectory field)

Stage 5 · Exfiltration — Covert SFTP Data Theft

MITRE Tactic	Technique ID	Technique Name
Exfiltration	T1048.002 T1059.004	Exfiltration Over Alternative Protocol Unix Shell

What happened

The attacker exfiltrated data via three SFTP sessions using a covert technique: instead of initiating a normal SFTP connection (which SSH logs would record), the attacker ran "bash -c /usr/lib/openssh/sftp-server" directly. This invokes the SFTP server binary from within a bash process, bypassing OpenSSH's session audit logging. Only Sysmon EventID 1 (process creation) captured it.

Key Evidence

- Session 1 (07:29:43): user tom, PID 51004, TerminalSession 6, LogonId 1002
- Session 2 (07:30:51): user pawbackup, PID 51004, TerminalSession 6, LogonId 1003
- Session 3: second pawbackup SFTP batch (separate process GUID)
- CommandLine: bash -c /usr/lib/openssh/sftp-server
- Process hash (bash):
SHA256=bc5945feb8bd26203ebfafea5ce1878bb2e32cb8fb50ab7ae395cfb1e1aaef1
- MITRE T1059.004: Unix shell used to launch sftp-server indirectly

Stage 6 · Ransomware Deployment — Scheduled Task Abuse

MITRE Tactic	Technique ID	Technique Name
Execution / Persistence	T1053.005 T1569.002	Scheduled Task Service Execution

What happened

The attacker used the shared local administrator account itadmin-tom to create a Windows Scheduled Task named "RunRansom" on all four user endpoints. The task ran C:\Users\Public\ransom.exe at the highest privilege level (/RL HIGHEST). Using a shared local admin account meant the same command worked on all four machines without needing separate credentials — a common lateral movement shortcut enabled by poor credential hygiene.

Key Evidence

- Command: schtasks /Create /TN RunRansom /TR "C:\Users\Public\ransom.exe" /SC ONCE /ST 00:00 /RL HIGHEST /RU itadmin-tom /F
- Launcher: cmd.exe /C schtasks [...] > C:\Windows\Temp\yuYeaQvr.tmp 2>&1
- Affected hosts: paw-tom (~07:52), paw-leo (~07:56), paw-marie (~08:00), paw-tabitha (~08:05)
- Shared local admin account itadmin-tom enabled automatic spread across all 4 hosts
- Temp file used: C:\Windows\Temp\yuYeaQvr.tmp (redirected schtasks output)

Stage 7 · Impact — Ransomware Execution & File Encryption

MITRE Tactic	Technique ID	Technique Name
Impact	T1486	Data Encrypted for Impact

What happened

Approximately 9 seconds after each scheduled task was created, ransom.exe executed and immediately spawned 30+ child threads per host. This multi-threaded approach is characteristic of BlackCat/ALPHV's Rust-based encryption engine, which parallelises file encryption across all available CPU cores to complete as fast as possible. Combined with the prior exfiltration (double extortion), the attacker can now demand ransom for both decryption and non-publication of stolen data.

Key Evidence

- Binary: C:\Users\Public\ransom.exe
- Execution: ~9 seconds after schtasks create event on each host
- 30+ child processes spawned simultaneously per host (encryption threads)
- All 4 user endpoints presumed fully encrypted
- Total C2 connection sessions logged: 16

Stage 8 · Anti-Forensics — Log Suppression

MITRE Tactic	Technique ID	Technique Name
Defense Evasion	T1562.002	Disable Windows Event Logging

What happened

Following the SFTP exfiltration sessions, all SSH authentication and Sysmon process logs from paw-backup ceased. This is an active log suppression technique: the attacker killed or reconfigured the syslog/journald pipeline, or manipulated Sysmon directly, to prevent forensic reconstruction of the lateral movement, the su to pawbackup, and any additional post-exploitation commands. Log gaps like this are a strong indicator of active attacker presence and should always be treated as suspicious.

Key Evidence

- Last paw-backup Sysmon event: 07:30:51 (SFTP Session 2)
- No further SSH auth or sysmon:linux events after 07:30:51 despite activity being inferred
- Log suppression masks: su to pawbackup, possible persistence mechanisms, cleanup commands
- Absence of logs is itself evidence of attacker action

4. Indicators of Compromise (IoCs)

4.1 Network Indicators

Type	Value	Port/Proto	Purpose
IP — Attacker	194.50.16.198	TCP/22 (SSH)	SSH brute-force source — Stage 1
IP — C2/SIEM	10.10.79.41:8989	TCP/8989	Splunk forwarder / C2 collection endpoint
IP — Cloud	10.10.50.10	TCP/80, TCP/443	Undocumented cloud asset — IMDS enum — Stage 2
IP — IMDS	169.254.169.254	TCP/80 (HTTP)	Azure IMDS — OAuth2 token endpoint, 22 hits
IP — External	4.208.26.197	TCP/443 (HTTPS)	External IP contacted post-IMDS — Stage 2
IP — GitHub	185.199.108.133	TCP/443 (HTTPS)	GitHub CDN — likely tool/payload download
Domain	WatchingCatsLounge.thm	N/A	Dark-web credential marketplace — pre-stage

4.2 Compromised Accounts

Account	System	How Abused
tom	paw-backup (Linux)	SSH brute-forced; used for recon (cat /etc/passwd) and SFTP Session 1
pawbackup	paw-backup (Linux)	Pivoted to via su; used for SFTP Sessions 2 & 3 (data exfiltration)
itadmin-tom	All Windows hosts	Shared local admin; used to create RunRansom schtask across 4 endpoints
leo	Dark web leak	Credential leaked — SSH rejected (invalid user on paw-backup)
marie	Dark web leak	Credential leaked — SSH rejected (invalid user on paw-backup)
tabitha	Dark web leak	Credential leaked — SSH rejected (invalid user on paw-backup)

4.3 File & Process Indicators

Type	Value	Context
Executable	C:\Users\Public\ransom.exe	Ransomware binary — dropped on all 4 Windows hosts
Executable	/usr/lib/openssh/sftp-server	Used for covert exfiltration via bash -c
Script	bash -c /usr/lib/openssh/sftp-server	Full command used for SFTP bypass exfil
Sched. Task	RunRansom	Malicious scheduled task name — all 4 hosts
Temp File	C:\Windows\Temp\yuYeaQvr.tmp	schtasks command stdout redirect file
SHA256	bc5945feb8bd26203ebfafea5ce1878bb2e32cb8fb50ab7ae395cfb1e1aaef1	Hash of /usr/bin/bash (sftp exfil process)
SHA256	dda0961715677dff3cd560e1933379c0cca73c0b6e19fef2737492ebc1de1700	Hash of /usr/bin/cat (recon process)

4.4 Affected Hosts

Host	IP	OS	Impact
paw-backup	10.10.79.41	Linux	Initial access, recon, lateral movement, exfil source; log suppression active
paw-tom	Unknown	Windows	Ransomware deployed; 30+ encryption threads; PRESUMED ENCRYPTED
paw-leo	Unknown	Windows	Ransomware deployed; 30+ encryption threads; PRESUMED ENCRYPTED
paw-marie	Unknown	Windows	Ransomware deployed; 30+ encryption threads; PRESUMED ENCRYPTED
paw-tabitha	Unknown	Windows	Ransomware deployed; 30+ encryption threads; PRESUMED ENCRYPTED
10.10.50.10	Cloud	Unknown	IMDS token theft; Azure credential compromise suspected

5. MITRE ATT&CK TTP Mapping

ID	Technique	Tactic	Evidence
T1078	Valid Accounts	Initial Access	Leaked credentials used for SSH and Windows schtask
T1078.003	Valid Accounts: Local Accounts	Lateral Movement	itadmin-tom shared local admin — ransomware spread
T1110.001	Brute Force: Password Guessing	Credential Access	SSH brute-force from 194.50.16.198 for all 4 accounts
T1552.005	Cloud Instance Metadata API	Credential Access	22 IMDS requests by tom from 10.10.50.10
T1087.001	Account Discovery: Local Account	Discovery	cat /etc/passwd from /home/tom
T1059.004	Unix Shell	Execution	bash -c /usr/lib/openssh/sftp-server (3 sessions)
T1048.002	Exfiltration Over Alternative Protocol	Exfiltration	Covert SFTP exfil bypassing SSH session logging
T1053.005	Scheduled Task	Execution	RunRansom schtask on all 4 Windows endpoints
T1569.002	System Services: Service Execution	Execution	ransom.exe triggered via scheduled task
T1486	Data Encrypted for Impact	Impact	30+ encryption threads per host × 4 hosts
T1562.002	Impair Defenses: Disable Event Logging	Defense Evasion	paw-backup logs suppressed after 07:30:51

6. Recommended Remediation Actions

6.1 Immediate (0–24 hours)

- **ISOLATE all four encrypted Windows endpoints (paw-tom, paw-leo, paw-marie, paw-tabitha) from the network NOW.**
- RESET passwords for tom, leo, marie, tabitha, pawbackup, itadmin-tom across all systems.
- BLOCK 194.50.16.198 at perimeter firewall. Monitor for new source IPs from same campaign.
- REVOKE all Azure OAuth2 tokens and managed-identity credentials associated with 10.10.50.10.
- TAKE forensic images of paw-backup before any remediation — logs will be needed for full IR.
- DELETE "RunRansom" scheduled task on any endpoint not yet executed (if any still reachable).
- IDENTIFY 10.10.50.10 — this asset is not in known inventory and must be located.

6.2 Short-Term (1–7 days)

- Eliminate shared local admin accounts. Deploy Microsoft LAPS or equivalent to break lateral movement.
- Enforce MFA on all SSH access. Implement Fail2Ban or DenyHosts on paw-backup.
- Restrict SFTP server binary execution with Linux AppArmor or SELinux profiles.
- Restore encrypted endpoints from verified clean backups — confirm backups are unaffected.
- Conduct full Azure IAM audit — rotate all service principals and managed identity tokens.
- Expand DarkSearch / dark-web monitoring to include all employee email variations.

6.3 Long-Term (1–4 weeks)

- Segment paw-backup into a dedicated VLAN with strict ACLs — backup servers must not be reachable from user endpoints.
- Deploy EDR on Linux hosts. Sysmon For Linux alone is insufficient for real-time response.
- Enable IMDS token binding / instance metadata service restrictions on all Azure VMs.
- Brief leadership: assume exfiltrated data is in attacker possession — prepare double-extortion response plan.
- Conduct organisation-wide password audit and mandatory password resets for all exposed accounts.

7. Evidences

2026-09-22 07:19:30.718	10.10.79.41:8989	tom	10.10.50.10	allowed
2026-09-22 07:19:31.718	10.10.79.41:8989	tom	127.0.0.53	allowed
2026-09-22 07:19:31.718	10.10.79.41:8989	tom	10.10.50.10	allowed
2026-09-22 07:20:45.718	10.10.79.41:8989	tom	10.10.50.10	allowed
2026-09-22 07:20:45.718	10.10.79.41:8989	tom	10.10.50.10	allowed
2026-09-22 07:20:45.718	10.10.79.41:8989	tom	10.10.50.10	allowed
2026-09-22 07:20:45.718	10.10.79.41:8989	tom	127.0.0.53	allowed
2026-09-22 07:20:46.718	10.10.79.41:8989	tom	10.10.50.10	allowed

time ↕	host ↕	✓	_raw ↕
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2026-09-22 07:17:56.718	10.10.79.41:8989		{"BOOT_ID": "f33fd83fee844cf0b38d5cc212fd38c4", "COMM": "ssh", "EXEC": "/usr/sbin/ssh", "GID": "0", "MACHINE_ID": "ec24dd5cf52366b9616da7794e90f24", "PID": "invalid user leo 194.50.16.198 port 60418 [preauth]", "host": "paw-backup", "index": "main", "source": "journal://sysmon", "timestamp": "09/22/2026 08:17:56.718", "vendor_produ
2026-09-22 07:17:56.718	10.10.79.41:8989		{"BOOT_ID": "f33fd83fee844cf0b38d5cc212fd38c4", "COMM": "ssh", "EXEC": "/usr/sbin/ssh", "GID": "0", "MACHINE_ID": "ec24dd5cf52366b9616da7794e90f24", "PID": "from 194.50.16.198 port 60418:11: Bye Bye [preauth]", "host": "paw-backup", "index": "main", "source": "journal://sysmon", "timestamp": "09/22/2026 08:17:56.718", "vendor_produ
2026-09-22 07:17:56.718	10.10.79.41:8989		{"_raw": "paw-backup sshd[1048]: Disconnected from invalid user leo 194.50.16.198 port 60418 [preauth]", "host": "paw-backup", "index": "main", "source": "journal://sysmon", "timestamp": "09/22/2026 08:17:56.718", "vendor_produ
2026-09-22 07:17:56.718	10.10.79.41:8989		{"_raw": "paw-backup sshd[1048]: Received disconnect from 194.50.16.198 port 60418:11: Bye Bye [preauth]", "host": "paw-backup", "index": "main", "source": "journal://sysmon", "timestamp": "09/22/2026 08:17:56.718", "vendor_produ
2026-09-22 07:17:56.718	10.10.79.41:8989		{"_raw": "paw-backup sshd[1048]: Invalid user leo from 194.50.16.198 port 60418", "host": "paw-backup", "index": "main", "source": "/var/log/auth.log", "timestamp": "09/22/2026 08:17:56.718", "vendor_produ
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dest_ip	dest_port	count
169.254.169.254	80	22
185.199.108.133	443	1
4.208.26.197	443	1

10.10.50.10	169.254.169.254	80	22
10.10.50.10	185.199.108.133	443	1
10.10.50.10	4.208.26.197	443	1

2026-09-22 07:29:43.718	10.10.79.41:8989	root	{ "BOOT_ID": "f33fd83fee844cf0b38d5cc212fd38c4", "COMM": "systemd", "CommandLine": "bash -c /usr/lib/openssh/sftp-server", "Company": "-", "Computer": "01fc615a0f97", "Hashes": "SHA256=bc5945febbdb26203ebfafe5ce1878bb2e32cb8f50ab7ae395cfeb1aaef1", "Image": "/usr/bin/bash", "IntegrityLevel": "no level", "PID": "885", "PRIORITY": "6", "ParentCommandLine": "-", "ParentImage": "-", "ParentProcessGUID": "{00000000-0000-0000-0000-000000000000}", "ParentShell": "SEQUENCE": "51432", "SEQUENCE_UID": "05e3c9389edf40b88e40b5362e75f4fb", "SHA256": "bc5945febbdb26203ebfafe5ce1878bb2e32cb8f50ab7ae395cfeb1aaef1", "SystemOperation": "EventSource\\Provider Name=\"Linux-Sysmon\" Guid='{ff032593-a8d3-4f13-b0d6-01fc615a0f97}'><EventID>{<Version>0</Version><SystemOperation/><Channel><Computer>paw-backup</Computer><Security UserID='0'></System><EventData><Data Name='RuleName'>TechniqueID Name='Operational'></Data><Data Name='bin/bash'></Data><Data Name='FileVersion'></Data><Data Name='Description'></Data><Data Name='Product'></Data><Data Name='3b85-6827-ea03-000000000000'></Data><Data Name='LoginID'>1002</Data><Data Name='TerminalSessionID'>6</Data><Data Name='IntegrityLevel'></Data><Data Name='ParentImage'></Data><Data Name='ParentCommandLine'></Data><Data Name='ParentUser'></Data><Data Name='EventData'><Events>, \"action\": \"000000000000\", \"parent_process_id\": \"51003\", \"parent_process_name\": \"-\", \"parent_process_path\": \"-\", \"process\": \"bash -c /usr/lib/openssh/sftp-s as95fecf0000\", \"process_hash\": \"SHA256=bc5945febbdb26203ebfafe5ce1878bb2e32cb8f50ab7ae395cfeb1aaef1\", \"process_id\": \"51004\", \"process_inti [process\", \"report\", \"user\": \"root\", \"user_id\": \"0\", \"vendor_product\": \"Sysmon For Linux\", \"source\": \"sysmon:linux\", \"timestamp\": \"09/22/2026
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2026-09-22 07:19:30.718		10.10.50.10	4.208.26.197		443
2026-09-22 07:19:31.718		10.10.50.10	185.199.108.133		443
2026-09-22 07:20:45.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:45.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:45.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254		80
2026-09-22 07:18:00.718	10.10.79.41:8989	{ "BOOT_ID": "f33fd83fee844cf0b38d5cc212fd38c4", "COMM": "ssh", "EXE": "/usr/sbin/ssh", "GID": "0", "MACHINE_ID": "ec24ddd5cf52366b9616da7794e90f24", "PID password for invalid user tabitha from 194.50.16.198 port 60434 ssh2", "host": "paw-backup", "index": "main", "source": "journal://sysmon", "timestamp": "09			
2026-09-22 07:18:00.718	10.10.79.41:8989	{ "BOOT_ID": "f33fd83fee844cf0b38d5cc212fd38c4", "COMM": "ssh", "EXE": "/usr/sbin/ssh", "GID": "0", "MACHINE_ID": "ec24ddd5cf52366b9616da7794e90f24", "PID password for invalid user leo from 194.50.16.198 port 60422 ssh2", "host": "paw-backup", "index": "main", "source": "journal://sysmon", "timestamp": "09			
2026-09-22 07:18:00.718	10.10.79.41:8989	{ "BOOT_ID": "f33fd83fee844cf0b38d5cc212fd38c4", "COMM": "ssh", "EXE": "/usr/sbin/ssh", "GID": "0", "MACHINE_ID": "ec24ddd5cf52366b9616da7794e90f24", "PID password for invalid user marie from 194.50.16.198 port 60428 ssh2", "host": "paw-backup", "index": "main", "source": "journal://sysmon", "timestamp": "09			
2026-09-22 07:18:00.718	10.10.79.41:8989	{ "_raw": "paw-backup sshd[1051]: Failed password for invalid user marie from 194.50.16.198 port 60428 ssh2", "host": "paw-backup", "index": "main", "so			
2026-09-22 07:20:45.718	10.10.79.41:8989	{ "Guid": "{ff032593-a8d3-4f13-b0d6-01fc615a0f97}", "Name": "Linux-Sysmon", "ProcessID": "885", "TechniqueID": "T1105", "TechniqueName": "Ingress", "ThreadID <Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime=\\\"\\\"/><EventRecordID>5692</EventRecordID><Correlation><Execution ProcessID=\\\"885\\\" Name=\\\"UtcTime\\\"></Data><Data Name=\\\"ProcessGuid\\\">{ec24ddd5-3c04-6827-8dfe-2ab121570000}</Data><Data Name=\\\"ProcessId\\\">4323</Data><Data Name=\\\" Name=\\\"SourceHostName\\\"></Data><Data Name=\\\"SourcePort\\\">33142</Data><Data Name=\\\"SourcePortName\\\"></Data><Data Name=\\\"DestinationIsIPv6\\\">false backup", "index": "main", "process": "sysmon", "source": "/var/log/syslog", "datasource": "syslog", "timestamp": "09/22/2026 08:20:45.718"}			
2026-09-22 07:20:45.718	10.10.79.41:8989	{ "BOOT_ID": "f33fd83fee844cf0b38d5cc212fd38c4", "COMM": "sysmon", "Computer": "paw-backup", "DestinationHostName": "-", "DestinationIp": "169.254.169.254" 01fc615a0f97)", "Image": "/usr/bin/curl", "Initiated": "true", "Keywords": "0x8000000000000000", "MACHINE_ID": "ec24ddd5cf52366b9616da7794e90f24", "Name": "Transfer", "SEQUENCE": "51206", "SEQUENCE_ID": "05e303989edf4b088e40b5362e75f4fb", "SYSTEMD_CGROUP": "/system.slice/sysmon.service", "SYSTEMD_UNIT": "sysmon.", "TRANSPORT": "syslog", "TechniqueID": "T1105", "TechniqueName": "Ingress", "ThreadID": "885", "UID": "0", "User": "tom", "UserId": "0", "_raw": "<Event><SystemTime=\\\"\\\"/><EventRecordID>5692</EventRecordID><Correlation><Execution ProcessID=\\\"885\\\" ThreadID=\\\"885\\\"><Channel>Linux-Sysmon/Operational 3c04-6827-8dfe-2ab121570000</Data><Data Name=\\\"ProcessId\\\">4323</Data><Data Name=\\\"Image\\\">/usr/bin/curl</Data><Data Name=\\\"User\\\">tom</Data><Data Name=\\\"SourcePortName\\\"></Data><Data Name=\\\"DestinationIsIPv6\\\">false</Data><Data Name=\\\"DestinationIp\\\">169.254.169.254</Data><Data Name=\\\"Dest: </Event>", "action": "allowed", "app": "/usr/bin/curl", "dest": "169.254.169.254", "dest_ip": "169.254.169.254", "dest_port": "80", "direction": "outbound", "id": "2ab121570000", "process_id": "4323", "process_name": "curl", "protocol": "IP", "protocol_version": "ipv4", "signature": "Network connection", "signature_id ["communicate", "network", "transport": "tcp", "transport_dest_port": "tcp/80", "user": "tom", "user_id": "0", "vendor_product": "Sysmon For Linux", "dataso			
2026-09-22 07:20:45.718	10.10.79.41:8989	{ "Guid": "{ff032593-a8d3-4f13-b0d6-01fc615a0f97}", "Name": "Linux-Sysmon", "ProcessID": "885", "TechniqueID": "T1105", "TechniqueName": "Ingress", "ThreadID <Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime=\\\"\\\"/><EventRecordID>5690</EventRecordID><Correlation><Execution ProcessID=\\\"885\\\" Name=\\\"UtcTime\\\"></Data><Data Name=\\\"ProcessGuid\\\">{ec24ddd5-3c04-6827-8dfe-54c75d5d0000}</Data><Data Name=\\\"ProcessId\\\">4274</Data><Data Name=\\\" Name=\\\"SourceHostName\\\"></Data><Data Name=\\\"SourcePort\\\">33126</Data><Data Name=\\\"SourcePortName\\\"></Data><Data Name=\\\"DestinationIsIPv6\\\">false backup", "index": "main", "process": "sysmon", "source": "/var/log/syslog", "datasource": "syslog", "timestamp": "09/22/2026 08:20:45.718"}			
2026-09-22 07:18:24.718				tom	
2026-09-22 07:19:30.718		10.10.50.10	4.208.26.197	tom	allowed
2026-09-22 07:19:31.718		127.0.0.53	127.0.0.1	tom	allowed
2026-09-22 07:19:31.718		10.10.50.10	185.199.108.133	tom	allowed
2026-09-22 07:20:45.718		10.10.50.10	169.254.169.254	tom	allowed
2026-09-22 07:20:45.718		10.10.50.10	169.254.169.254	tom	allowed
2026-09-22 07:20:45.718		10.10.50.10	169.254.169.254	tom	allowed
2026-09-22 07:20:45.718		127.0.0.53	127.0.0.1	tom	allowed
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254	tom	allowed
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254	tom	allowed
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254	tom	allowed
2026-09-22 07:20:46.718		10.10.50.10	169.254.169.254	tom	allowed
2026-09-22 07:16:35.718					root
2026-09-22 07:16:35.718					root
2026-09-22 07:16:36.718					root
2026-09-22 07:16:37.718					root
2026-09-22 07:16:37.718					root

Appendix A — Splunk Queries (Verified Working)

All queries below returned results in the PawPressMe Splunk environment and are confirmed against log evidence in this report.

A1. Dark Web Credential Leak Alert

Retrieve the DarkSearch alert that triggered the credential leak investigation.

```
index=* sourcetype="DarkSearchMe" organization="pawpress.me"
| table _time, affected_account_email, category, severity, leak_source
| sort _time
```

A2. SSH Brute Force — paw-backup

Find all SSH failed/disconnected events originating from the attacker IP 194.50.16.198.

```
index=* "paw-backup" "sshd" "Failed password"
| table _time, host, _raw
| sort _time
```

A3. All Activity by Compromised Accounts

Full account activity log for the four leaked employee accounts across all systems.

```
index=* (user="leo@pawpress.me" OR user="marie@pawpress.me" OR
user="tabitha@pawpress.me" OR user="tom@pawpress.me"
OR user="leo" OR user="marie" OR user="tabitha" OR user="tom")
| table _time, host, user, src_ip, action, EventCode
| sort _time
```

A4. Account tom Network Activity (Includes IMDS Requests)

All network connections attributed to user tom — includes IMDS enumeration from 10.10.50.10.

```
index=* (user="tom" OR user="tom@pawpress.me")
| table _time, src_ip, dest_ip, user, action
| sort _time
```

A5. IMDS Enumeration — 10.10.50.10

Network connections from the unknown cloud asset to IMDS and external IPs.

```
index=* src_ip="10.10.50.10" dest_ip IN
("169.254.169.254","185.199.108.133","4.208.26.197")
| table _time, src_ip, dest_ip, dest_port, bytes_out
| sort _time
```

A6. Full Traffic Profile — 10.10.50.x Subnet

Aggregated destination summary for all cloud asset activity.

```
index=* src_ip="10.10.50.*"  
| table _time, src_ip, dest_ip, dest_port, user  
| stats count by src_ip, dest_ip, dest_port  
| sort -count
```

A7. IMDS Requests (All Sources)

Catch any host querying the Azure IMDS endpoint — useful for detecting additional compromised assets.

```
index=main  
| search _raw="*169.254.169.254*" OR _raw="*metadata/identity*"   
| table _time, host, _raw  
| sort _time
```

A8. Covert SFTP Exfiltration Events

Identify bash-launched sftp-server processes used to bypass SSH session logging on paw-backup.

```
index=* "sftp-server" OR "sftp"  
| table _time, host, user, _raw  
| sort _time
```

A9. Ransomware — Scheduled Task Creation & Execution

Find all RunRansom scheduled task creation and ransom.exe execution events across Windows hosts.

```
index=* "194.50.16.198"  
| table _time, host, _raw  
| sort _time
```

For ransomware execution events specifically:

```
index=* "ransom.exe" OR "RunRansom"  
| table _time, host, _raw  
| sort _time
```